Subject: Mathematics **CBSE CLASS 10 – CHAPTER TEST (SET – 5)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 7: Coordinate Geometry** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided wherever specified.
2. **Section A** comprises Multiple Choice / Objective Questions of 1 mark each.
3. **Section B** comprises Short Answer Type questions.
4. **Section C** comprises Long Answer & Competency Based questions.
5. Use of calculators or electronic devices is strictly prohibited. Draw neat figures wherever required.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. In the following figure, P and Q are points of trisection of line segment AB :

A ——— P ——— Q ——— B

the value of $\frac{AB}{PB}$ = [1]

(A) 1 (B) 1.5 (C) $\frac{2}{3}$ (D) 2

Q2. If the points $A(1,2), O(0,0)$ and $C(a,b)$ are collinear, then: [1]

(A) $a = b$ (B) $a = 2b$ (C) $2a = b$ (D) $a = -b$

Q3. **Assertion [1]

(A) :** The point $(-2,4)$ divides the line segment joining the points $(-4,8)$ and $(5,-10)$ in the ratio $2:7$ internally. **Reason (R):** If three points P, Q and R are collinear, then $PQ + QR = PR$. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

Q4. The coordinates of the midpoint of the line segment joining the points $(-5,7)$ and $(-1,3)$ are: [1]

(A) $(-3,5)$ (B) $(3,5)$ (C) $(-3,-5)$ (D) $(5,-3)$

Q5. If $(\frac{a}{3}, 4)$ is the midpoint of the line segment joining the points $Q(-6,5)$ and $R(-2,3)$, then the value of a is: [1]

(A) -4 (B) -12 (C) 12 (D) -6

SECTION B: Short Answer Type Questions (6 Marks)

Q6. Calculate the ratio in which the line segment joining the points $A(6,5)$ and $B(4,-3)$ is divided by the line $y = 2$. Also, find the point of intersection. [3]

Q7. If $P(9a-2, -b)$ divides the line segment joining $A(3a+1, -3)$ and $B(8a, 5)$ in the ratio $3:1$, find the values of a and b. [3]

SECTION C: Long Answer & Competency Based Questions (9 Marks)

Q8. Find the coordinates of the points of trisection of the line segment joining the points $P(5, -4)$ and $Q(-4, 2)$. ¿ **OR** ¿ (b) Find the ratio in which the point $P(6, k)$ divides the line segment joining the points $M(4, 2)$ and $N(8, -4)$. Hence find the value of k. ¿ ¿ [5]

Q9. Competency Case Study: A municipal corporation maps three community utility centers situated at points $P(2, 3)$, $Q(6, 7)$ and $R(8, 9)$ on a city plan. [4]

(i) Find the distance between utility center P and center Q.

(ii) Find the coordinates of a service station S located midway between P and R.

(iii) Check whether points P, Q and R lie on a straight road (are collinear) by comparing distance sums.

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